Indoor navigation: building review

Sam Ibrahim Building, UTSC | Six floors | Original photograph + official classroom cross-checks

Main finding: compare the building from top to bottom, but create access only where each floor has a real doorway or landing. Repeated seating or stair geometry can be a projection of a lower space.

The Cloud is the clearest example. Two east-side opening candidates appear on Level 2. On Level 3 the same seating footprint is unlabelled and bounded by a continuous east edge. The adjacent stair is outside that boundary. A descent to lower circulation is plausible; a direct doorway into the Cloud is not established.

Deliverable: a hard-codeable annotation inventory with stable IDs, normalized coordinates, original-pixel references, candidate portals, named core matches and explicit unresolved links. It is not a complete operational routing graph.

Important correction: the right-hand tiered room is the Conveyor, not the Connector. Official sources identify IA2150 as Conveyor; the current classroom table uses IA2150A. [S6]

Scope limit: small enclosures and faint openings are candidates, not verified building totals. The photograph cannot resolve every doorway or current access condition. Bboxes are review extents, never walkable polygons.

6
floor panels

366
feature regions

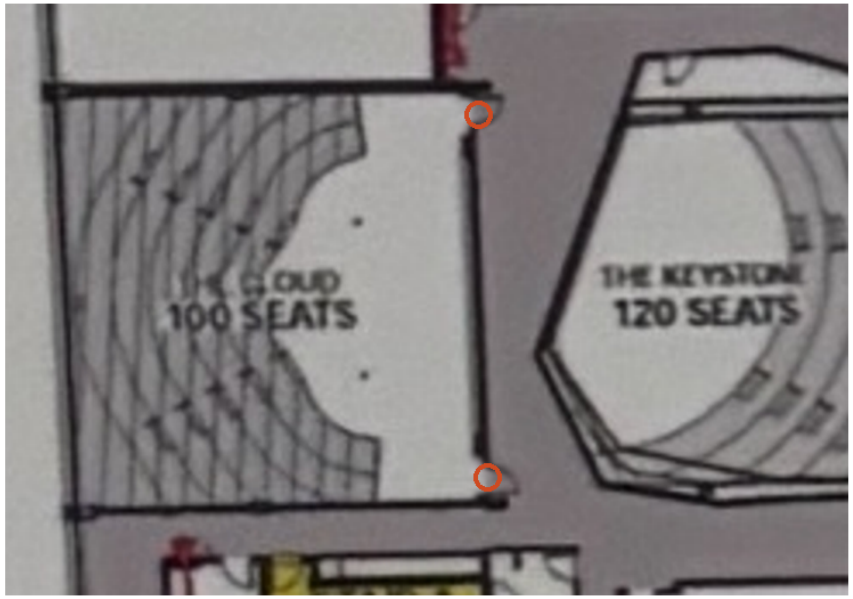
330
portal candidates

0
verified cross-floor edges

Cloud: same footprint, different access

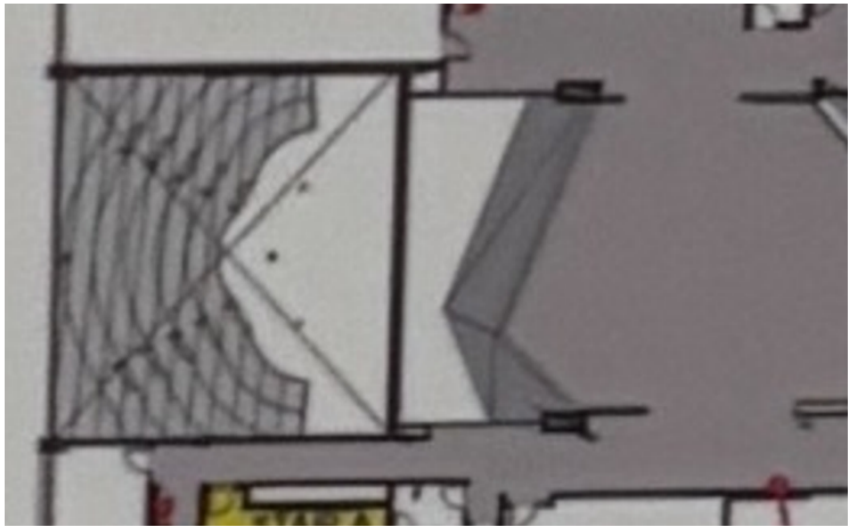
Original image pixels with deterministic markers; no AI reconstruction

L2 - original photograph crop



Two east-side opening candidates

L3 - original photograph crop



Continuous east boundary; no copied doors

Same room footprint is plausible. Stair endpoints remain unresolved.

Orange rings indicate two apparent Level 2 openings. Their absence from the corresponding Level 3 boundary supports a projected lower-room interpretation. This does not by itself prove an open void: a section or site observation is still needed.

Official cross-check: Cloud is IA2040, a tiered classroom, and the classroom inventory lists two entrances/exits. That supports the doorway count, not their exact elevation or current access. [S2, S3]

Level 5 | Uppermost drawn level

105 feature regions; 95 doorway/exit candidates. Synthetic labels are review IDs.



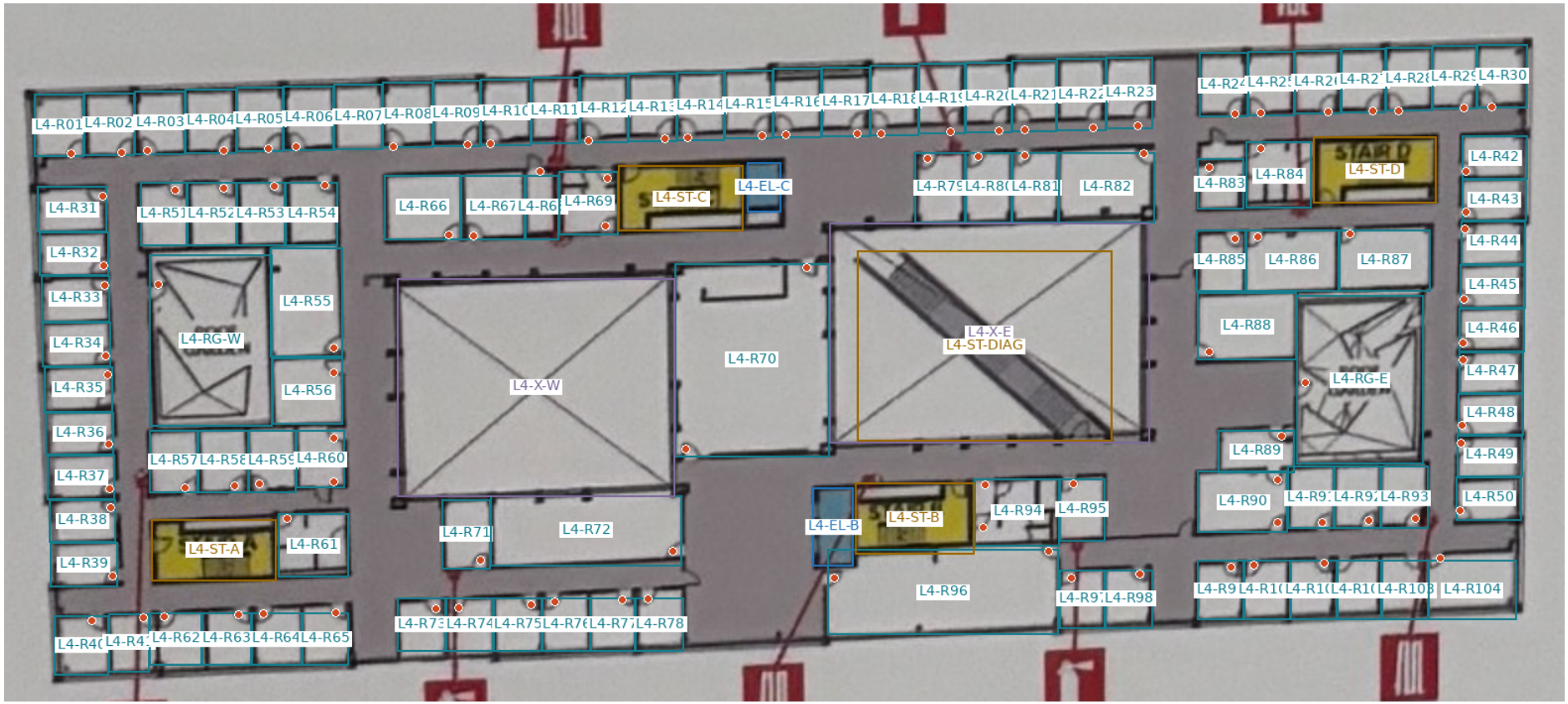
Observed: Trace the grey perimeter and internal corridor branches around the four large crossed regions. Numerous small white enclosures have door-swing candidates. Stair cores A-D and blue shafts beside B/C repeat. No room numbers are readable.

Connection review: The two side positions that are roof gardens on L4 appear as crossed regions here. Do not inherit garden floors or entrances onto L5. Keep the two central crossed regions excluded pending confirmation.

Overlay key: teal = room/enclosure; amber = stair feature; blue = elevator shaft; purple = uncertain/excluded region; orange dots = candidate openings. Boxes may overlap and are not floor boundaries.

Level 4 | Office-like rooms and roof gardens

115 feature regions; 108 doorway/exit candidates. Synthetic labels are review IDs.



Observed: Two labelled roof gardens flank the two large central crossed regions. The repetitive small-room pattern provides the clearest doorway vocabulary. All four named stair cores and both elevator shafts remain visible.

Connection review: The diagonal band through the central-right crossed region is a stair candidate. Its footprint crosses a non-floor candidate; only an identified stair surface and verified endpoints may be routed. Garden entrance/access remains unresolved.

Overlay key: teal = room/enclosure; amber = stair feature; blue = elevator shaft; purple = uncertain/excluded region; orange dots = candidate openings. Boxes may overlap and are not floor boundaries.

Level 3 | Projected rooms and changing circulation

41 feature regions; 35 doorway/exit candidates. Synthetic labels are review IDs.



Observed: Upper and lower grey circulation bands flank the central volume. Cloud seating repeats at left without the L2 label or two east-side opening symbols. The Conveyor repeats at right with changed access geometry.

Connection review: Record Cloud as a lower-room/projection candidate. Keep the stair immediately east of Cloud separate from the room. A route down to L2 is plausible but the destination landing is not proven. Do not copy L2 doors upward.

Overlay key: teal = room/enclosure; amber = stair feature; blue = elevator shaft; purple = uncertain/excluded region; orange dots = candidate openings. Boxes may overlap and are not floor boundaries.

Level 2 | Named teaching rooms and tiered interiors

39 feature regions; 36 doorway/exit candidates. Synthetic labels are review IDs.



Observed: Named rooms include Corner Shop, Spot, Mission Control, Portal, Living Room, Cloud, Keystone, Conveyor, Stage, Boundary and Rhythm. Cloud has two east-edge opening candidates. The Keystone has a separate tiered-room shape.

Connection review: Keep every room entrance distinct, including multiple doors into the same room. Preserve both user-identified central stair features as stairs, while leaving direction and landing identities unresolved. Do not treat seating rows as universal routes.

Overlay key: teal = room/enclosure; amber = stair feature; blue = elevator shaft; purple = uncertain/excluded region; orange dots = candidate openings. Boxes may overlap and are not floor boundaries.

Level 1 | Main lobby and exterior exit groups

34 feature regions; 29 doorway/exit candidates. Synthetic labels are review IDs.



Observed: The broad grey lobby links small teaching rooms and stair/elevator cores. Green exit arrows mark several perimeter exit groups. The posted YOU ARE HERE marker belongs to this level near the upper-right core.

Connection review: The large left white region can depict a different-height space; it is not automatically open lobby floor. Twelve exit review points represent arrow/threshold observations in groups, not twelve verified doors. Confirm exterior thresholds and vestibules.

Overlay key: teal = room/enclosure; amber = stair feature; blue = elevator shaft; purple = uncertain/excluded region; orange dots = candidate openings. Boxes may overlap and are not floor boundaries.

Level B | Lower auditoriums and service rooms

32 feature regions; 27 doorway/exit candidates. Synthetic labels are review IDs.



Observed: Arrow Group Innovation Hall is at left; Collaboratorium is central; a further tiered room is at right. Grey circulation surrounds these volumes. Four named stair cores plus an extra northwest core and two blue shafts are visible.

Connection review: The extra stair label is unclear. Repeated auditorium rows and sawtooth/wavy room-boundary details are not each doors. Separate electrical/mechanical areas from public routes; confirm the right-hand room relationship to L1.

Overlay key: teal = room/enclosure; amber = stair feature; blue = elevator shaft; purple = uncertain/excluded region; orange dots = candidate openings. Boxes may overlap and are not floor boundaries.

Connect landings, not overlapping shapes

Implementation model and the difference between seeing a feature and accessing it

1. Separate depiction from physical entity - A room can appear in more than one floor drawing. Store a physical room identity separately from its per-floor depiction. Cloud L3 is not automatically another classroom or another entrance.

2. Preserve individual doorways - Each doorway gets its own ID and a space on each side. A room with two entrances must not become a single point that teleports between doors. Add a verified internal path, including stairs, when needed.

3. Model stair endpoints - Give every stair its actual landing nodes. Each landing needs a floor and, for split levels, an elevation or unresolved elevation ID. A staircase may connect a room terrace, another floor, or both.

4. Use core matching only as an identity clue - A-D and the blue B/C-adjacent shafts repeat across all six drawings. Candidate adjacent-floor matches are supplied. Confirm door/lobby positions, served floors and re-entry before creating a traversable edge.

5. Respect drawing boundaries - Grey/mauve continuity supports circulation. A black boundary, void, railing or change of height overrides colour similarity. Neither line density nor grey shade establishes up/down direction.

6. Keep original evidence - Every marker traces to an original crop and source-photo coordinate. Review the uncertain records in place. Do not infer geometry from the illustrative image generated earlier in the conversation.

Official room cross-checks

20 teaching-room entries, 34 listed entrances/exits; not a building-wide door count

IA-B1010	1 entrance/exit	IA2060	2 entrances/exits
IA-B1050	2 entrances/exits	IA2100	1 entrance/exit
IA-B1100	2 entrances/exits	IA2110	1 entrance/exit
IA1120	1 entrance/exit	IA2120	1 entrance/exit
IA1150	3 entrances/exits	IA2150A	3 entrances/exits
IA1160	1 entrance/exit	IA2160	2 entrances/exits
IA2010	1 entrance/exit	IA2170	2 entrances/exits
IA2021	2 entrances/exits	IA3050	2 entrances/exits
IA2040	2 entrances/exits	IA3060	1 entrance/exit
IA2050	2 entrances/exits	IA3120	2 entrances/exits

Source [S2]. Counts are a completeness check, not positional evidence. Do not match a synthetic enclosure to a room number solely by its seat count. The photograph and current inventory can disagree, and the Conveyor naming/suffix needs reconciliation.

The original image lacks readable room numbers for most enclosures, especially on Levels 4 and 5. They are deliberately stored with synthetic IDs. Current access and individual entrance accessibility require separate checks.

Review order and implementation handoff

Use the JSON to populate a map editor; resolve topology before enabling navigation

Q01 At the Level 3 Cloud boundary, is there a door or only a continuous wall/guard edge?

Q02 Trace both landings of the open stair immediately east of the Cloud on Level 3. Does it reach Level 2 circulation?

Q03 For each Conveyor entrance, record the floor label and whether it reaches a room landing or through route.

Q04 Verify the endpoints of central open stairs and the Level 4 diagonal band independently.

Q05 Confirm the exact elevator door/lobby position beside cores B and C on each floor.

Q06 Confirm stair A-D served floors, entrance doors and re-entry restrictions.

Q07 Identify the extra yellow stair above C on B and Level 1.

Q08 Match upper-floor synthetic enclosure IDs to room signs and check each small doorway.

Q09 Confirm both Level 4 roof-garden entrances; do not inherit them on Level 5.

Q10 Measure two independent distances per floor and doorway clear widths where needed.

Level 5 | region inventory

Rows 1-12 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R01	North perimeter 1 (synthetic)	room candidate	0.224, 0.132	P001
L5-R02	North perimeter 2 (synthetic)	room candidate	0.261, 0.130	P002
L5-R03	North perimeter 3 (synthetic)	room candidate	0.306, 0.129	P003
L5-R04	North perimeter 4 (synthetic)	room candidate	0.376, 0.126	P004
L5-R05	North perimeter 5 (synthetic)	room candidate	0.431, 0.122	P005
L5-R06	North perimeter 6 (synthetic)	room candidate	0.477, 0.121	P006
L5-R07	North perimeter 7 (synthetic)	room candidate	0.530, 0.118	P007
L5-R08	North perimeter 8 (synthetic)	room candidate	0.567, 0.116	P008
L5-R09	North perimeter 9 (synthetic)	room candidate	0.598, 0.114	P009
L5-R10	North perimeter 10 (synthetic)	room candidate	0.630, 0.113	P010
L5-R11	North perimeter 11 (synthetic)	room candidate	0.660, 0.112	P011
L5-R12	North perimeter 12 (synthetic)	room candidate	0.690, 0.110	P012

Level 5 | region inventory

Rows 13-24 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R13	North perimeter 13 (synthetic)	room candidate	0.719, 0.109	P013
L5-R14	North perimeter 14 (synthetic)	room candidate	0.749, 0.107	P014
L5-R15	North perimeter 15 (synthetic)	room candidate	0.780, 0.106	P015
L5-R16	North perimeter 16 (synthetic)	room candidate	0.819, 0.105	none resolved
L5-R17	North perimeter 17 (synthetic)	room candidate	0.866, 0.102	P016
L5-R18	North perimeter 18 (synthetic)	room candidate	0.904, 0.100	P017
L5-R19	North perimeter 19 (synthetic)	room candidate	0.933, 0.099	P018
L5-R20	North perimeter 20 (synthetic)	room candidate	0.957, 0.119	P019
L5-R21	West perimeter 1 (synthetic)	room candidate	0.048, 0.128	P020
L5-R22	West perimeter 2 (synthetic)	room candidate	0.048, 0.196	P021
L5-R23	West perimeter 3 (synthetic)	room candidate	0.047, 0.262	P022
L5-R24	West perimeter 4 (synthetic)	room candidate	0.046, 0.327	P023

Level 5 | region inventory

Rows 25-36 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R25	West perimeter 5 (synthetic)	room candidate	0.045, 0.392	P024
L5-R26	West perimeter 6 (synthetic)	room candidate	0.045, 0.458	P025
L5-R27	West perimeter 7 (synthetic)	room candidate	0.044, 0.523	P026
L5-R28	West perimeter 8 (synthetic)	room candidate	0.043, 0.588	P027
L5-R29	West perimeter 9 (synthetic)	room candidate	0.043, 0.652	P028
L5-R30	West perimeter 10 (synthetic)	room candidate	0.042, 0.717	P029
L5-R31	West perimeter 11 (synthetic)	room candidate	0.041, 0.782	P030
L5-R32	West perimeter 12 (synthetic)	room candidate	0.041, 0.847	P031
L5-R33	West perimeter 13 (synthetic)	room candidate	0.040, 0.912	P032
L5-R34	East perimeter 1 (synthetic)	room candidate	0.955, 0.217	P033
L5-R35	East perimeter 2 (synthetic)	room candidate	0.953, 0.280	P034
L5-R36	East perimeter 3 (synthetic)	room candidate	0.952, 0.342	P035

Level 5 | region inventory

Rows 37-48 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R37	East perimeter 4 (synthetic)	room candidate	0.950, 0.405	P036
L5-R38	East perimeter 5 (synthetic)	room candidate	0.949, 0.469	P037
L5-R39	East perimeter 6 (synthetic)	room candidate	0.947, 0.533	P038
L5-R40	East perimeter 7 (synthetic)	room candidate	0.945, 0.597	P039
L5-R41	East perimeter 8 (synthetic)	room candidate	0.944, 0.660	P040
L5-R42	East perimeter 9 (synthetic)	room candidate	0.942, 0.723	P041
L5-R43	North of west exclusion 1 (synthetic)	room candidate	0.105, 0.272	P042
L5-R44	North of west exclusion 2 (synthetic)	room candidate	0.135, 0.272	P043
L5-R45	North of west exclusion 3 (synthetic)	room candidate	0.166, 0.272	P044
L5-R46	Small partitioned service enclosure by west inner block	service suite candidate	0.202, 0.248	P045
L5-R47	East of west exclusion	room candidate	0.192, 0.556	none resolved
L5-R48	South of west exclusion 1 (synthetic)	room candidate	0.107, 0.633	P046

Level 5 | region inventory

Rows 49-60 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R49	South of west exclusion 2 (synthetic)	room candidate	0.142, 0.633	P047
L5-R50	South of west exclusion 3 (synthetic)	room candidate	0.164, 0.633	P048
L5-R51	South of west exclusion 4 (synthetic)	room candidate	0.196, 0.633	P049
L5-R52	Service enclosure east of Stair A	service suite candidate	0.186, 0.759	P050, P051, P052
L5-R53	South of Stair A 1 (synthetic)	room candidate	0.094, 0.851	P053
L5-R54	South of Stair A 2 (synthetic)	room candidate	0.120, 0.851	P054
L5-R55	South of Stair A 3 (synthetic)	room candidate	0.146, 0.851	P055
L5-R56	South of Stair A 4 (synthetic)	room candidate	0.173, 0.851	P056
L5-R57	South of Stair A 5 (synthetic)	room candidate	0.199, 0.851	P057
L5-R58	West of Stair C 1 (synthetic)	room candidate	0.265, 0.264	P058
L5-R59	West of Stair C 2 (synthetic)	room candidate	0.301, 0.264	P059
L5-R60	West of Stair C 3 (synthetic)	room candidate	0.323, 0.264	P060

Level 5 | region inventory

Rows 61-72 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R61	West of Stair C 4 (synthetic)	room candidate	0.353, 0.264	P061
L5-R62	North of central east exclusion 1 (synthetic)	room candidate	0.582, 0.252	P062, P066
L5-R63	North of central east exclusion 2 (synthetic)	room candidate	0.654, 0.252	P063
L5-R64	North of central east exclusion 3 (synthetic)	room candidate	0.685, 0.252	P064
L5-R65	North of central east exclusion 4 (synthetic)	room candidate	0.731, 0.252	none resolved
L5-R66	North of central east exclusion 5 (synthetic)	room candidate	0.777, 0.252	P065
L5-R67	Service enclosure west of Stair D	service suite candidate	0.814, 0.241	none resolved
L5-R68	Isolated central enclosure	room candidate	0.484, 0.462	P067
L5-R69	North of east exclusion 1 (synthetic)	room candidate	0.799, 0.372	P068
L5-R70	North of east exclusion 2 (synthetic)	room candidate	0.833, 0.372	P069
L5-R71	North of east exclusion 3 (synthetic)	room candidate	0.878, 0.372	P070
L5-R72	West of east exclusion	room candidate	0.796, 0.463	P071, P072

Level 5 | region inventory

Rows 73-84 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R73	South of east exclusion 1 (synthetic)	room candidate	0.788, 0.719	P073
L5-R74	South of east exclusion 2 (synthetic)	room candidate	0.823, 0.719	P074
L5-R75	South of east exclusion 3 (synthetic)	room candidate	0.854, 0.719	P075
L5-R76	South of east exclusion 4 (synthetic)	room candidate	0.883, 0.719	P076
L5-R77	Southeast perimeter 1 (synthetic)	room candidate	0.758, 0.849	none resolved
L5-R78	Southeast perimeter 2 (synthetic)	room candidate	0.787, 0.849	P077
L5-R79	Southeast perimeter 3 (synthetic)	room candidate	0.818, 0.849	P078
L5-R80	Southeast perimeter 4 (synthetic)	room candidate	0.848, 0.849	P079
L5-R81	Southeast perimeter 5 (synthetic)	room candidate	0.878, 0.849	P080
L5-R82	Southwest central suite west enclosure	room candidate	0.266, 0.777	P081, P082
L5-R83	Southwest central suite internal circulation	suite circulation candidate	0.371, 0.765	P083
L5-R84	Southwest central suite upper small room	room candidate	0.388, 0.739	P084

Level 5 | region inventory

Rows 85-96 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-R85	Southwest central suite upper east room	room candidate	0.425, 0.725	P085
L5-R86	Southwest central suite bottom row 1 (synthetic)	room candidate	0.325, 0.881	P086
L5-R87	Southwest central suite bottom row 2 (synthetic)	room candidate	0.369, 0.881	P087
L5-R88	Southwest central suite bottom row 3 (synthetic)	room candidate	0.399, 0.881	P088
L5-R89	Southwest central suite bottom row 4 (synthetic)	room candidate	0.430, 0.881	P089
L5-R90	Immediately below Stair B	room candidate	0.536, 0.807	P090
L5-R91	South of Stair B lower enclosure	room candidate	0.535, 0.870	P091
L5-R92	Southeast of Stair B lower enclosure	room candidate	0.610, 0.842	P092
L5-R93	Large enclosure east of Stair B service block	room candidate	0.701, 0.786	P093
L5-R94	Service enclosure east of Stair B	service suite candidate	0.637, 0.728	P094, P095
L5-ST-A	STAIR A	stair shaft	0.121, 0.763	none resolved
L5-ST-B	STAIR B	stair shaft	0.567, 0.730	none resolved

Level 5 | region inventory

Rows 97-105 of 105. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L5-ST-C	STAIR C	stair shaft	0.430, 0.258	none resolved
L5-ST-D	STAIR D	stair shaft	0.878, 0.236	none resolved
L5-EL-C	Elevator shaft beside Stair C	elevator shaft	0.483, 0.241	none resolved
L5-EL-B	Elevator shaft beside Stair B	elevator shaft	0.514, 0.745	none resolved
L5-X1	Crossed exclusion 1	void candidate	0.128, 0.453	none resolved
L5-X2	Crossed exclusion 2	void candidate	0.331, 0.527	none resolved
L5-X3	Crossed exclusion 3	void candidate	0.621, 0.466	none resolved
L5-X4	Crossed exclusion 4	void candidate	0.857, 0.541	none resolved
L5-X5	Crossed exclusion 5	void candidate	0.265, 0.893	none resolved

Level 4 | region inventory

Rows 1-12 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R01	North perimeter west segment 1 (synthetic)	room candidate	0.035, 0.175	P001
L4-R02	North perimeter west segment 2 (synthetic)	room candidate	0.067, 0.173	P002
L4-R03	North perimeter west segment 3 (synthetic)	room candidate	0.100, 0.171	P003
L4-R04	North perimeter west segment 4 (synthetic)	room candidate	0.132, 0.169	P004
L4-R05	North perimeter west segment 5 (synthetic)	room candidate	0.164, 0.167	P005
L4-R06	North perimeter west segment 6 (synthetic)	room candidate	0.195, 0.165	P006
L4-R07	North perimeter west segment 7 (synthetic)	room candidate	0.227, 0.162	none resolved
L4-R08	North perimeter west segment 8 (synthetic)	room candidate	0.259, 0.160	P007
L4-R09	North perimeter west segment 9 (synthetic)	room candidate	0.290, 0.158	P008
L4-R10	North perimeter west segment 10 (synthetic)	room candidate	0.321, 0.156	P009
L4-R11	North perimeter west segment 11 (synthetic)	room candidate	0.353, 0.154	none resolved
L4-R12	North perimeter west segment 12 (synthetic)	room candidate	0.384, 0.152	P010

Level 4 | region inventory

Rows 13-24 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R13	North perimeter west segment 13 (synthetic)	room candidate	0.415, 0.149	P011
L4-R14	North perimeter west segment 14 (synthetic)	room candidate	0.446, 0.147	P012
L4-R15	North perimeter west segment 15 (synthetic)	room candidate	0.477, 0.145	P013
L4-R16	North perimeter west segment 16 (synthetic)	room candidate	0.508, 0.143	P014
L4-R17	North perimeter west segment 17 (synthetic)	room candidate	0.539, 0.141	P015
L4-R18	North perimeter west segment 18 (synthetic)	room candidate	0.571, 0.139	P016
L4-R19	North perimeter west segment 19 (synthetic)	room candidate	0.601, 0.137	P017
L4-R20	North perimeter west segment 20 (synthetic)	room candidate	0.631, 0.135	P018
L4-R21	North perimeter west segment 21 (synthetic)	room candidate	0.660, 0.133	P019
L4-R22	North perimeter west segment 22 (synthetic)	room candidate	0.691, 0.131	P020
L4-R23	North perimeter west segment 23 (synthetic)	room candidate	0.721, 0.129	P021
L4-R24	North perimeter east segment 1 (synthetic)	room candidate	0.782, 0.117	P022

Level 4 | region inventory

Rows 25-36 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R25	North perimeter east segment 2 (synthetic)	room candidate	0.812, 0.115	P023
L4-R26	North perimeter east segment 3 (synthetic)	room candidate	0.842, 0.113	P024
L4-R27	North perimeter east segment 4 (synthetic)	room candidate	0.871, 0.111	P025
L4-R28	North perimeter east segment 5 (synthetic)	room candidate	0.901, 0.109	P026
L4-R29	North perimeter east segment 6 (synthetic)	room candidate	0.930, 0.106	P027
L4-R30	North perimeter east segment 7 (synthetic)	room candidate	0.960, 0.104	P028
L4-R31	West perimeter 1 (synthetic)	room candidate	0.044, 0.295	P029
L4-R32	West perimeter 2 (synthetic)	room candidate	0.045, 0.359	P030
L4-R33	West perimeter 3 (synthetic)	room candidate	0.046, 0.423	P031
L4-R34	West perimeter 4 (synthetic)	room candidate	0.047, 0.488	P032
L4-R35	West perimeter 5 (synthetic)	room candidate	0.047, 0.553	P033
L4-R36	West perimeter 6 (synthetic)	room candidate	0.048, 0.616	P034

Level 4 | region inventory

Rows 37-48 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R37	West perimeter 7 (synthetic)	room candidate	0.049, 0.678	P035
L4-R38	West perimeter 8 (synthetic)	room candidate	0.050, 0.741	P036
L4-R39	West perimeter 9 (synthetic)	room candidate	0.051, 0.803	P037
L4-R40	Southwest corner enclosure	room candidate	0.050, 0.920	P038
L4-R41	Narrow southwest bottom enclosure	room candidate	0.080, 0.915	P039
L4-R42	East perimeter 1 (synthetic)	room candidate	0.955, 0.222	P040
L4-R43	East perimeter 2 (synthetic)	room candidate	0.955, 0.282	P041
L4-R44	East perimeter 3 (synthetic)	room candidate	0.954, 0.342	P042
L4-R45	East perimeter 4 (synthetic)	room candidate	0.954, 0.405	P043
L4-R46	East perimeter 5 (synthetic)	room candidate	0.953, 0.467	P044
L4-R47	East perimeter 6 (synthetic)	room candidate	0.953, 0.528	P045
L4-R48	East perimeter 7 (synthetic)	room candidate	0.952, 0.588	P046

Level 4 | region inventory

Rows 49-60 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R49	East perimeter 8 (synthetic)	room candidate	0.952, 0.648	P047
L4-R50	East perimeter 9 (synthetic)	room candidate	0.952, 0.709	P048
L4-R51	North of west roof garden 1 (synthetic)	room candidate	0.103, 0.303	P049
L4-R52	North of west roof garden 2 (synthetic)	room candidate	0.133, 0.303	P050
L4-R53	North of west roof garden 3 (synthetic)	room candidate	0.164, 0.303	P051
L4-R54	North of west roof garden 4 (synthetic)	room candidate	0.197, 0.303	P052
L4-R55	East of west roof garden upper enclosure	room candidate	0.194, 0.429	P053
L4-R56	East of west roof garden lower enclosure	room candidate	0.195, 0.556	P054
L4-R57	South of west roof garden 1 (synthetic)	room candidate	0.109, 0.656	P055
L4-R58	South of west roof garden 2 (synthetic)	room candidate	0.140, 0.656	P056
L4-R59	South of west roof garden 3 (synthetic)	room candidate	0.172, 0.656	P057
L4-R60	Small divided enclosure southeast of west roof garden	service suite candidate	0.203, 0.653	P058, P059

Level 4 | region inventory

Rows 61-72 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R61	Service enclosure east of Stair A	service suite candidate	0.198, 0.777	P060
L4-R62	South of Stair A 1 (synthetic)	room candidate	0.110, 0.911	P061
L4-R63	South of Stair A 2 (synthetic)	room candidate	0.142, 0.911	P062
L4-R64	South of Stair A 3 (synthetic)	room candidate	0.174, 0.911	P063
L4-R65	South of Stair A 4 (synthetic)	room candidate	0.206, 0.911	P064
L4-R66	West of Stair C 1 (synthetic)	room candidate	0.268, 0.292	P065
L4-R67	West of Stair C 2 (synthetic)	room candidate	0.314, 0.292	P066
L4-R68	West of Stair C 3 (synthetic)	room candidate	0.345, 0.292	P067
L4-R69	Service enclosure west of Stair C	service suite candidate	0.375, 0.286	P068, P069
L4-R70	Central enclosure between crossed regions	room candidate	0.479, 0.511	P070, P071
L4-R71	South of west central exclusion small room	room candidate	0.296, 0.760	P072
L4-R72	South of west central exclusion large room	room candidate	0.373, 0.755	P073

Level 4 | region inventory

Rows 73-84 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R73	South central perimeter 1 (synthetic)	room candidate	0.267, 0.890	P074
L4-R74	South central perimeter 2 (synthetic)	room candidate	0.299, 0.890	P075
L4-R75	South central perimeter 3 (synthetic)	room candidate	0.329, 0.890	P076
L4-R76	South central perimeter 4 (synthetic)	room candidate	0.361, 0.890	P077
L4-R77	South central perimeter 5 (synthetic)	room candidate	0.390, 0.890	P078
L4-R78	South central perimeter 6 (synthetic)	room candidate	0.420, 0.890	P079
L4-R79	North of east central exclusion 1 (synthetic)	room candidate	0.600, 0.264	P080
L4-R80	North of east central exclusion 2 (synthetic)	room candidate	0.631, 0.264	P081
L4-R81	North of east central exclusion 3 (synthetic)	room candidate	0.661, 0.264	P082
L4-R82	North of east central exclusion 4 (synthetic)	room candidate	0.706, 0.264	P083
L4-R83	Small enclosure west of Stair D service suite	room candidate	0.780, 0.259	P084
L4-R84	Service enclosure west of Stair D	service suite candidate	0.817, 0.246	P085

Level 4 | region inventory

Rows 85-96 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R85	North of east roof garden 1 (synthetic)	room candidate	0.780, 0.369	P086
L4-R86	North of east roof garden 2 (synthetic)	room candidate	0.825, 0.369	P087
L4-R87	North of east roof garden 3 (synthetic)	room candidate	0.884, 0.369	P088
L4-R88	Northwest of east roof garden	room candidate	0.796, 0.463	P089
L4-R89	Southwest of east roof garden upper small room	room candidate	0.803, 0.641	P090
L4-R90	Southwest of east roof garden lower room	room candidate	0.794, 0.714	P091, P092
L4-R91	South of east roof garden 1 (synthetic)	room candidate	0.838, 0.708	P093
L4-R92	South of east roof garden 2 (synthetic)	room candidate	0.867, 0.708	P094
L4-R93	South of east roof garden 3 (synthetic)	room candidate	0.897, 0.708	P095
L4-R94	Service enclosure east of Stair B	service suite candidate	0.649, 0.729	P096, P097
L4-R95	Small room east of Stair B service suite	room candidate	0.691, 0.725	P098
L4-R96	Large south enclosure below Stair B	room candidate	0.602, 0.843	P099, P100

Level 4 | region inventory

Rows 97-108 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-R97	South perimeter west of east corridor 1 (synthetic)	room candidate	0.691, 0.853	P101
L4-R98	South perimeter west of east corridor 2 (synthetic)	room candidate	0.721, 0.853	P102
L4-R99	Southeast perimeter 1 (synthetic)	room candidate	0.780, 0.840	P103
L4-R100	Southeast perimeter 2 (synthetic)	room candidate	0.809, 0.840	P104
L4-R101	Southeast perimeter 3 (synthetic)	room candidate	0.839, 0.840	P105
L4-R102	Southeast perimeter 4 (synthetic)	room candidate	0.868, 0.840	none resolved
L4-R103	Southeast perimeter 5 (synthetic)	room candidate	0.898, 0.840	none resolved
L4-R104	Southeast perimeter 6 (synthetic)	room candidate	0.941, 0.840	P106
L4-ST-A	STAIR A	stair shaft	0.135, 0.783	none resolved
L4-ST-B	STAIR B	stair shaft	0.583, 0.738	none resolved
L4-ST-C	STAIR C	stair shaft	0.434, 0.280	none resolved
L4-ST-D	STAIR D	stair shaft	0.878, 0.239	none resolved

Level 4 | region inventory

Rows 109-115 of 115. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L4-EL-C	Elevator shaft beside Stair C	elevator shaft	0.487, 0.263	none resolved
L4-EL-B	Elevator shaft beside Stair B	elevator shaft	0.531, 0.750	none resolved
L4-RG-W	ROOF GARDEN	roof garden	0.132, 0.483	P107
L4-RG-E	ROOF GARDEN	roof garden	0.869, 0.540	P108
L4-X-W	Central crossed exclusion W	void candidate	0.341, 0.550	none resolved
L4-X-E	Central crossed exclusion E	void candidate	0.632, 0.473	none resolved
L4-ST-DIAG	Diagonal stair-like band inside central-east crossed region	stair candidate	0.628, 0.491	none resolved

Level 3 | region inventory

Rows 1-12 of 41. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L3-R01	The Hive	classroom	0.102, 0.200	P01, P02
L3-R02	Maples (lettering uncertain)	classroom	0.241, 0.162	P03
L3-R03	Unlabelled room north of Stair C	room candidate	0.395, 0.122	P04
L3-R04	Unlabelled small upper room	room candidate	0.517, 0.090	P05
L3-R05	Unlabelled narrow upper room	room candidate	0.578, 0.172	P06
L3-R06	The … Shop (60 seats; name uncertain)	classroom	0.701, 0.170	P07, P08
L3-R07	The Cloud footprint shown from L3	room projection unresolved	0.089, 0.512	none resolved
L3-R08	The Conveyor	tiered classroom	0.810, 0.493	P09, P10, P11
L3-R09	Unlabelled southwest compartment	room candidate	0.037, 0.755	P12
L3-R10	Unlabelled room south/west of Stair A	room candidate	0.062, 0.884	P13
L3-R11	Unlabelled room south/east of Stair A	room candidate	0.153, 0.882	P14
L3-R12	Unlabelled south room 1	room candidate	0.242, 0.841	P15

Level 3 | region inventory

Rows 13-24 of 41. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L3-R13	Unlabelled south room 2	room candidate	0.335, 0.838	P16
L3-R14	Unlabelled south small compartment 1	room candidate	0.393, 0.774	P17
L3-R15	Unlabelled south small compartment 2	room candidate	0.423, 0.774	P18
L3-R16	Unlabelled south small room	room candidate	0.410, 0.881	P19
L3-R17	Unlabelled room south of Stair B	room candidate	0.572, 0.872	P20
L3-R18	Unlabelled southeast room 1	room candidate	0.722, 0.830	P21, P22
L3-R19	Unlabelled southeast room 2	room candidate	0.873, 0.825	P23, P24
L3-SA	Stair A	stair shaft	0.118, 0.769	P32
L3-SB	Stair B	stair shaft	0.558, 0.756	P33
L3-SC	Stair C	stair shaft	0.423, 0.235	P34
L3-SD	Stair D	stair shaft	0.871, 0.226	P35
L3-E01	Elevator shaft beside Stair C	elevator shaft	0.477, 0.233	none resolved

Level 3 | region inventory

Rows 25-36 of 41. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L3-E02	Elevator shaft beside Stair B	elevator shaft	0.506, 0.769	none resolved
L3-U01	Core A east support rooms	service space unresolved	0.172, 0.768	P25
L3-U02	Core C west support rooms	service space unresolved	0.342, 0.248	P26, P27
L3-U03	Core D west support rooms	service space unresolved	0.816, 0.237	P28
L3-U04	Core D east support room	service space unresolved	0.924, 0.237	P29
L3-U05	Core B east support rooms	service space unresolved	0.624, 0.753	P30, P31
L3-C01	Upper corridor	corridor candidate	0.553, 0.327	none resolved
L3-C02	Lower corridor	corridor candidate	0.502, 0.692	none resolved
L3-C03	Central grey circulation/terrace	corridor candidate	0.291, 0.504	none resolved
L3-C04	East perimeter corridor	corridor candidate	0.937, 0.496	none resolved
L3-C05	North central recess	corridor candidate	0.518, 0.241	none resolved
L3-C06	Northeast circulation	corridor candidate	0.882, 0.118	none resolved

Level 3 | region inventory

Rows 37-41 of 41. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L3-C07	South central recess	corridor candidate	0.468, 0.838	none resolved
L3-OS01	Stair feature immediately east of Cloud	open stair candidate	0.215, 0.515	none resolved
L3-OS02	Stair feature east side of central terrace	open stair user confirmed	0.395, 0.505	none resolved
L3-OS03	Northeast central diagonal feature	open stair or projection candidate	0.585, 0.457	none resolved
L3-V01	Central white region	void or other level projection candidate	0.541, 0.502	none resolved

Level 2 | region inventory

Rows 1-12 of 39. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L2-R01	The Corner Shop	classroom	0.092, 0.193	P01
L2-R02	The Spot	classroom	0.239, 0.158	P02
L2-R03	Mission Control	classroom	0.585, 0.153	P03
L2-R04	The Portal	classroom	0.679, 0.152	P04
L2-R05	The Living Room	classroom	0.755, 0.152	P05
L2-R06	The Cloud	tiered classroom	0.098, 0.516	P06, P07
L2-R07	The Keystone	tiered classroom	0.299, 0.491	P08, P09
L2-R08	The Conveyor	tiered classroom	0.803, 0.491	P10, P11
L2-R09	The Stage	classroom	0.282, 0.848	P12
L2-R10	Boundary	classroom	0.755, 0.811	P13, P14
L2-R11	Rhythm	classroom	0.903, 0.802	P15, P16
L2-R12	Unlabelled room west of central south space	room candidate	0.378, 0.808	P17, P18

Level 2 | region inventory

Rows 13-24 of 39. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L2-R13	Unlabelled central south space	room candidate	0.456, 0.824	P19, P20
L2-R14	Unlabelled space south of Stair B	room candidate	0.561, 0.879	P21
L2-SA	Stair A	stair shaft	0.133, 0.776	P33
L2-SB	Stair B	stair shaft	0.588, 0.751	P34
L2-SC	Stair C	stair shaft	0.437, 0.231	P35
L2-SD	Stair D	stair shaft	0.883, 0.220	P36
L2-E01	Elevator shaft beside Stair C	elevator shaft	0.489, 0.229	none resolved
L2-E02	Elevator shaft beside Stair B	elevator shaft	0.537, 0.766	none resolved
L2-U01	Core A west support rooms	service space unresolved	0.075, 0.776	P22, P23
L2-U02	Core A east support rooms	service space unresolved	0.189, 0.774	P24
L2-U03	Core C west support rooms	service space unresolved	0.361, 0.229	P25, P26
L2-U04	Core D west support rooms	service space unresolved	0.830, 0.235	P27

Level 2 | region inventory

Rows 25-36 of 39. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L2-U05	Core D east support rooms	service space unresolved	0.937, 0.221	P28, P29
L2-U06	Core B east support rooms	service space unresolved	0.660, 0.814	P30, P31, P32
L2-V01	Crossed area east of Stage	excluded unresolved	0.378, 0.912	none resolved
L2-C01	Upper corridor	corridor candidate	0.563, 0.305	none resolved
L2-C02	Lower corridor	corridor candidate	0.502, 0.697	none resolved
L2-C03	Cloud–Keystone passage	corridor candidate	0.196, 0.511	none resolved
L2-C04	Central diagonal circulation	corridor candidate	0.475, 0.645	none resolved
L2-C05	East central circulation	corridor candidate	0.660, 0.521	none resolved
L2-C06	East perimeter corridor	corridor candidate	0.966, 0.495	none resolved
L2-C07	North central circulation	corridor candidate	0.414, 0.121	none resolved
L2-C08	Northeast circulation	corridor candidate	0.891, 0.114	none resolved
L2-C09	Southwest circulation	corridor candidate	0.124, 0.892	none resolved

Level 2 | region inventory

Rows 37-39 of 39. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L2-OS01	Central open stair group	open stair user confirmed	0.574, 0.499	none resolved
L2-OS02	Stair candidate southeast of Keystone	open stair candidate	0.419, 0.583	none resolved
L2-V02	Central white region	void or other level projection candidate	0.506, 0.499	none resolved

Level 1 | region inventory

Rows 1-12 of 34. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L1-R01	Brainstorm	room candidate	0.702, 0.235	P01
L1-R02	Dark Matter	room candidate	0.553, 0.553	P02
L1-R03	Right tiered classroom; name unreadable	room candidate	0.747, 0.644	none resolved
L1-R04	Central small enclosure west of Dark Matter	room candidate	0.502, 0.581	P03
L1-R05	White enclosure above Stair A	room candidate	0.175, 0.691	P04
L1-R06	Small enclosure east of Stair A	room candidate	0.215, 0.785	P05
L1-R07	Washroom-like enclosure below Stair B	room candidate	0.576, 0.846	none resolved
L1-R08	Ancillary enclosure east of Stair B	room candidate	0.607, 0.766	none resolved
L1-R09	Ancillary enclosure west of upper stair cores	room candidate	0.384, 0.258	P06
L1-R10	Ancillary enclosure east of upper stair	room candidate	0.493, 0.187	P07
L1-R11	Vestibule or service enclosure west of Stair D	room candidate	0.777, 0.247	P08, X04
L1-R12	Right classroom northwest ancillary enclosure	room candidate	0.667, 0.438	P09

Level 1 | region inventory

Rows 13-24 of 34. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L1-R13	Right classroom west-upper vestibule	room candidate	0.652, 0.485	P10
L1-R14	Right classroom west-middle ancillary enclosure	room candidate	0.636, 0.560	none resolved
L1-R15	Right classroom west-lower vestibule	room candidate	0.620, 0.633	P11
L1-U01	Large left white region above basement auditorium footprint	unknown region	0.200, 0.438	none resolved
L1-U02	Angular feature within left white region	unknown region	0.258, 0.512	none resolved
L1-U03	White curved region beneath Stair C	unknown region	0.439, 0.360	none resolved
L1-SA	Stair A	stair shaft	0.130, 0.735	P12, X11
L1-SB	Stair B	stair shaft	0.548, 0.818	P13, X10
L1-SC	Stair C	stair shaft	0.435, 0.284	P14
L1-SD	Stair D	stair shaft	0.829, 0.244	P15, X05
L1-SEXTRA	STAIR_EXTRA_NW	stair shaft	0.431, 0.188	X03
L1-EC	Blue elevator shaft beside Stair C	elevator shaft	0.481, 0.274	P16

Level 1 | region inventory

Rows 25-34 of 34. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
L1-EB	Blue elevator shaft beside Stair B	elevator shaft	0.504, 0.776	P17
L1-OS01	Open stair candidate below Stair C	open stair candidate	0.434, 0.467	none resolved
L1-OS02	Open stair candidate northwest lobby	open stair candidate	0.295, 0.287	none resolved
L1-OS03	Open stair candidate near Stair A	open stair candidate	0.264, 0.767	none resolved
L1-OS04	Narrow steps beside central enclosure	open stair candidate	0.479, 0.597	none resolved
L1-OS05	Step groups on east classroom boundary	open stair candidate	0.835, 0.605	none resolved
L1-C01	Central Level 1 lobby	circulation candidate	0.387, 0.719	X01, X02, X08, X09
L1-C02	Upper Level 1 circulation	circulation candidate	0.585, 0.276	none resolved
L1-C03	Right Level 1 circulation	circulation candidate	0.828, 0.421	X06, X07
L1-C04	Southwest Level 1 circulation	circulation candidate	0.210, 0.846	X12

Level B | region inventory

Rows 1-12 of 32. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
LB-R01	Arrow Group Innovation Hall	room candidate	0.183, 0.454	P01, P02, P03, P04
LB-R02	The Collaboratorium	room candidate	0.643, 0.438	P05, P06
LB-R03	The Green Room	room candidate	0.453, 0.869	P07
LB-R04	Right tiered classroom; name unreadable	room candidate	0.846, 0.712	P08, P09
LB-R05	Main Electrical	room candidate	0.865, 0.266	P10
LB-R06	Mechanical & Sprinkler Pump	room candidate	0.938, 0.749	P11
LB-R07	Unlabelled service area around Stair A	room candidate	0.136, 0.872	none resolved
LB-R08	Unlabelled washroom-like enclosure	room candidate	0.270, 0.875	P12
LB-R09	Small lower service enclosure west	room candidate	0.341, 0.917	P13
LB-R10	Small lower service enclosure east	room candidate	0.374, 0.917	none resolved
LB-R11	Small enclosure north of Stair B	room candidate	0.601, 0.766	none resolved
LB-R12	Unlabelled enclosure east of Stair B	room candidate	0.693, 0.807	P14

Level B | region inventory

Rows 13-24 of 32. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
LB-R13	Unlabelled enclosure southwest of Main Electrical	room candidate	0.770, 0.474	P15
LB-R14	Small enclosure below LB-R13	room candidate	0.755, 0.613	none resolved
LB-R15	Triangular enclosure beside right classroom	room candidate	0.833, 0.488	none resolved
LB-R16	Unlabelled enclosure east/south of Stair D	room candidate	0.954, 0.361	P16
LB-R17	Unlabelled triangular service enclosure	room candidate	0.964, 0.488	P17
LB-R18	Small enclosure west of Stair C	room candidate	0.341, 0.247	P18
LB-U01	White curved region beneath Stair C	unknown region	0.462, 0.361	P19
LB-U02	Ancillary area northeast of upper stair core	unknown region	0.488, 0.180	P20
LB-U03	Upper peripheral strip	unknown region	0.320, 0.076	none resolved
LB-SA	Stair A	stair shaft	0.128, 0.832	P21, P22
LB-SB	Stair B	stair shaft	0.611, 0.833	P23
LB-SC	Stair C	stair shaft	0.434, 0.271	P24

Level B | region inventory

Rows 25-32 of 32. Coordinates are crop-relative fractions. Full evidence and portal coordinates are in JSON.

ID	Label	Feature type	Centre x, y	Related portal IDs
LB-SD	Stair D	stair shaft	0.921, 0.296	P25
LB-SEXTRA	STAIR_EXTRA_NW	stair shaft	0.410, 0.175	none resolved
LB-EC	Blue elevator shaft beside Stair C	elevator shaft	0.507, 0.261	P26
LB-EB	Blue elevator shaft beside Stair B	elevator shaft	0.558, 0.843	P27
LB-OS01	Possible open steps below Stair C	open stair candidate	0.416, 0.404	none resolved
LB-C01	Central basement circulation	circulation candidate	0.443, 0.590	none resolved
LB-C02	Lower basement circulation	circulation candidate	0.686, 0.927	none resolved
LB-C03	Right diagonal circulation	circulation candidate	0.838, 0.637	none resolved

Sources, provenance and practical limits

External sources corroborate room identities and counts, not a complete route graph

[S1] User photograph - All six floor drawings and legend; no architectural survey precision.

source/original_poster.jpg

[S2] UTSC classroom inventory - Teaching-room door totals and types; does not locate the doors or establish entrance elevations.

https://www.utsc.utoronto.ca/iits/classroom-support?field_room_location_value=IA

[S3] UTSC Cloud room description - Cloud is IA2040, tiered, with curved seating; linked virtual tour available on source page.

https://www.utsc.utoronto.ca/iits/cloud-sam-ibrahim-building-ia2040

[S4] UTSC IA2021 room details - Tiered room; two entrances/exits.

https://www.utsc.utoronto.ca/iits/ia2021

[S5] UTSC SDG Data Challenge - Official event names Keystone as IA2021.

https://www.utsc.utoronto.ca/cms/events/utsc-sdg-data-challenge

[S6] UTSC safety committee minutes - Names IA2150 as Conveyor. Exact relationship to current IA2150A inventory suffix requires checking.

https://www.utsc.utoronto.ca/ehs/sites/utsc.utoronto.ca.ehs/files/docs/Minutes%20March%202025%20final.pdf

[S7] ZAS architect description - Interlocking multilevel spaces, embedded seating and terraces; no exact stair endpoints.

https://www.zasa.com/news/-utsc-inaugurates-sam-ibrahim-building

[S8] UTSC Arrow Innovation Hall - Name maps to IAB1050.

https://utsc.utoronto.ca/iits/arrow-innovation-hall-sam-ibrahim-building-iab1050

[S9] UTSC Collaboratorium - Name maps to IAB1100.

https://utsc.utoronto.ca/iits/collaboratorium-sam-ibrahim-building-iab1100

Verification performed: unique IDs; normalized coordinate bounds; ordered boxes; valid portal references; named-core membership across six depictions; visual review of annotated floor sheets. These checks do not establish physical connectivity, completeness or a metric scale.